

Related Work: Solar Panel Thermal Hotspot Detection

Comparison of existing research/projects similar to our AI-based thermal hotspot detection system

S.No	Related Project / Reference	Approach Used	Advantages	Disadvantages	Remark (Relevance to Our Project)
1	Towards Efficient Solar Panel Inspection: A YOLO-based Method for Hotspot Detection (IEEE, 2024)	Integrates a thermal sensor with a YOLO deep learning model to detect PV panel defects from thermal images	Directly validates a YOLO-based approach for hotspot detection; good accuracy on standard datasets	Uses higher-resolution thermal cameras than our low-cost MLX90640 sensor, so results aren't directly comparable	Closest match to our core idea (thermal + YOLO); confirms our overall approach is sound
2	Deep Learning-Driven Thermal Imaging Hotspot Detection using YOLOv10 (drone-based PV arrays)	Fine-tunes YOLOv10 on drone-captured thermal images of large solar farms for real-time hotspot detection	Real-time detection; large, varied training dataset from real solar plants	Built for drone/utility-scale farms with expensive thermal cameras — not designed for a low-cost handheld unit like ours	Shows our newer, lighter YOLOv8-nano choice is reasonable for our smaller-scale, low-power use case
3	HOTSPOT-YOLO: Lightweight Attention-Driven Model for Drone-Based PV Inspection (arXiv, 2025)	YOLOv11 with an EfficientNet backbone and attention mechanism, optimized for small/subtle thermal anomalies	High accuracy (~90.8% mAP); lightweight and built for real-time, resource-constrained use	Still targets drone-based large-scale inspection, and the attention-enhanced model is more complex to implement than YOLOv8-nano	Good reference for future improvement – attention mechanisms could improve our detection accuracy later
4	Two-Tier Semantic Segmentation (2TSS) Framework for Hotspot Localization (UAV thermal images)	Two-stage segmentation: first separates panel from background, then localizes the hotspot region within it	High interpretability; compares multiple ML/DL models, useful benchmarking data	Segmentation-based, not detection-based — heavier computation, less suited to a low-power Raspberry Pi	Confirms detection (YOLO) is a lighter, more practical choice than segmentation for our embedded hardware
5	ST-YOLO: PV Defect Detection using Infrared Thermal Imaging and Machine Vision	Attention Residual Network with multi-scale convolution branch to detect small-scale thermal anomalies	Handles small/hard-to-see anomalies well; addresses class imbalance in fault datasets	Requires more computational resources than a lightweight nano model can offer	Useful reference for handling small/early-stage hotspots that our system may currently miss
6	Edge-Guided Mask R-CNN for PV Hotspot Thermal Image Segmentation	Uses an edge-guided feature pyramid with Mask R-CNN to precisely outline hotspot boundaries	Very precise hotspot boundary detection, useful for severity estimation	Mask R-CNN is computationally heavy not practical for real-time inference on a Raspberry Pi 4	-Reinforces why we chose a lightweight detector (YOLOv8-nano) over heavier segmentation models
7	AP-YOLOv5: Improved Anchors & Prediction Heads for PV Hotspot Detection	Modifies YOLOv5's anchor boxes and prediction heads specifically for small hotspot regions on panels	87.8% mAP with strong recall; anchor tuning improves detection of small hotspots specifically	Based on the older YOLOv5 architecture; anchor-tuning adds design complexity we haven't attempted	Suggests a possible future upgrade path — tuning anchors for small hotspots could improve our model further